Dissertation Defense: Essays in Bayesian Inference and Deep Learning

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Bayesian Inference for Gamma Models

- We propose a framework of data augmentation strategy which can be applied to a variety of models:
  - Negative binomial regression for count data; Topic models in natural language processing; Generalized Pareto distribution in extreme value analysis, ...
- What have I done since proposal:
  - Renamed the distribution: Exponential Reciprocal Gamma (ERG).
  - Discussed sampling methods for ERG.
  - Unified gamma models under a general probability density.
  - Developed the general version of MCMC and EM algorithm.
  - Showed how to extend the framework and simplify the algorithms.
  - Completed the simulation study.

ERG(0) Distribution

Definition

Random variable X_0 has an exponential reciprocal gamma distribution, ERG(0), if and only if its density function $p_0(x)$ satisfies the following identity

$$E\left(e^{-s^2 X_0}\right) = \int_0^\infty e^{-s^2 x} p_0(x) dx = \frac{e^{-\gamma s}}{\Gamma(1+s)}, \quad s > 0.$$

where $\gamma = -\psi(1) \approx 0.57721$ is the Euler-Mascheroni constant and $\psi(s) = \frac{d}{ds} \log \Gamma(s)$ is the digamma function.

Remark

The product representation for the reciprocal gamma function due to Weierstrass is,

$$\frac{e^{-\gamma s}}{\Gamma(1+s)} = \prod_{k=1}^{\infty} \left(1 + \frac{s}{k}\right) e^{-\frac{s}{k}}, \; s \in \mathbb{C}/\{0, -1, -2, ...\}$$

Some distribution results in literature

Hartman (1976) proves the existence of a distribution V_t, for $0 < t \leq e^{-\gamma}$, such that

$$E\left(e^{-s^2 V_t}\right) = \frac{t^s}{\Gamma(1+s)}.$$

Roynette and Yor (2005) prove the existence of an infinitely divisible distribution H_a^Γ, with density function $p_a^\Gamma(x)$, such that for $a > 0$

$$E\left(e^{-\frac{1}{2}s^2 H_a^\Gamma}\right) = \int_0^\infty e^{-\frac{1}{2}s^2 x} p_a^\Gamma(x)dx = \frac{\Gamma(a)}{\Gamma(a+s)} e^{\psi(a)s}$$

which follows from the general product representation for the reciprocal gamma function,

$$\frac{\Gamma(a)}{\Gamma(a+s)} e^{\psi(a)s} = \prod_{k=0}^{\infty} \left(1 + \frac{s}{a+k}\right) e^{-\frac{s}{a+k}}.$$

ERG(0) is a special case with $a = 1$. ERG(0) $\stackrel{d}{=} \frac{1}{2} H_1^\Gamma$.

General ERG(c) Distribution

Definition

The ERG(c) distribution is constructed as an exponential tilting of the ERG(0) density. Its density function is

$$p_c(x) = Z_c \cdot \exp\left(-c^2 x\right) p_0(x),\ x, c > 0.$$

Similarly, the integral identity of ERG(c) is given by

$$\begin{aligned} E\left(e^{-s^2 X_c}\right) &= \prod_{k=1}^{\infty} \left(\frac{k+\sqrt{s^2+c^2}}{k+c}\right) e^{-\frac{\sqrt{s^2+c^2}-c}{k}} \\ &= \frac{\Gamma(1+c)}{\Gamma(1+\sqrt{s^2+c^2})} e^{-\gamma(\sqrt{s^2+c^2}-c)} \end{aligned}$$

Relation to Generalized Inverse Gaussian

Theorem

The ERG(c) class of distributions can be constructed as an infinite sum of independent generalized inverse Gaussian (GIG) distributions as follows

$$ERG(c) \stackrel{d}{=} \sum_{k=1}^{\infty} GIG\left(-\frac{3}{2}, 2c^2, \frac{1}{2k^2}\right).$$

In particular, when $c = 0$, the GIG distribution reduces to inverse gamma distribution. Hence,

$$ERG(0) \stackrel{d}{=} \sum_{k=1}^{\infty} \frac{1}{4k^2} \Gamma_k^{-1}.$$

where Γ_k^{-1} are i.i.d. inverse gamma random variables with shape $\frac{3}{2}$ and scale 1.

Mean and Variance

Theorem

If G_k is a GIG random variable with $p = -\frac{3}{2}, a = 2c^2, b = \frac{1}{2k^2}$, then the mean and variance of the tail infinite sum $\sum_{k=N}^{\infty} G_k$ are

$$E\left(\sum_{k=N}^{\infty} G_k\right) = \frac{1}{2c}\left(\psi(N+c) - \psi(N)\right)$$

$$Var\left(\sum_{k=N}^{\infty} G_k\right) = \frac{1}{4c^3}\left(\psi(N+c) - \psi(N) - c\psi'(N+c)\right).$$

Setting $N = 1$ gives us the first two moments of ERG(c).

This theorem is useful for the generation ERG variables, construction of EM algorithm as well as approximation of Gibbs sampler.

Data Augmentation

The idea of the data augmentation strategy is to introduce a group of auxiliary random variables $\boldsymbol{\omega} = (\omega_1, ..., \omega_n)^T$ such that for $\boldsymbol{\omega} \in \Omega$,

$$p(x) = \int_{\Omega} p(x, \boldsymbol{\omega}) d\boldsymbol{\omega}$$

and the joint density $p(x, \boldsymbol{\omega})$ after augmentation is easier to deal with, as it doesn't consist of gamma functions any longer.

	Integral Representation	Auxiliary Variables
$\Gamma(\alpha)$	$\int_0^\infty x^{\alpha-1} e^{-x} dx$	Gamma
$\frac{1}{\Gamma(\alpha)}$	$\int_0^\infty \alpha e^{-\alpha^2 x + \gamma\alpha} p_0(x) dx$	ERG
$\frac{\Gamma(\alpha)}{\Gamma(\alpha+\beta)}$	$\int_0^1 \frac{1}{\Gamma(\beta)} x^{\alpha-1}(1-x)^{\beta-1} dx$	Beta

Goal

We unify the (posterior) densities of many disparate gamma problems

$$p(x) \propto \left(\prod_{\ell=1}^{L} \Gamma(g_\ell(x))\right)\left(\prod_{m=1}^{M} \frac{1}{\Gamma(h_m(x))}\right)\left(\prod_{n=1}^{N} \frac{\Gamma(j_n(x))}{\Gamma(j_n(x)+\beta_n)}\right) \cdot x^{p-1} e^{-ax^2+bx}$$

- The support is $(0, \infty)$.
- g, h, j are increasing linear functions.
- $M \geq 1$ and $a \geq 0$ and assume the density is integrable.
- g, h, j, β can depend on data.

By exploiting the integral identities related to gamma function, we derive a general data augmentation strategy for the above density.

$$p(x) = C \cdot \left(\prod_{\ell=1}^{L} \Gamma(g_\ell(x))\right)\left(\prod_{m=1}^{M} \frac{1}{\Gamma(h_m(x))}\right)\left(\prod_{n=1}^{N} \frac{\Gamma(j_n(x))}{\Gamma(j_n(x)+\beta_n)}\right) \cdot x^{p-1} e^{-ax^2+bx}$$

$$= \int C \cdot G(x,\boldsymbol{\tau}) \cdot H(x,\boldsymbol{\omega}) \cdot J(x,\boldsymbol{\eta}) \cdot Q(x) \cdot e^{-ax^2+bx} d\boldsymbol{\tau} d\boldsymbol{\omega} d\boldsymbol{\eta}$$

where

$$G(x,\boldsymbol{\tau}) = \prod_{\ell=1}^{L} \tau_\ell^{g_\ell(x)-1} e^{-\tau_\ell}$$

$$H(x,\boldsymbol{\omega}) = \exp\left\{-\sum_{m=1}^{M} h_m(x)^2 \omega_m + \gamma \sum_{m=1}^{M} h_m(x)\right\} \prod_{m=1}^{M} p_0(\omega_m)$$

$$J(x,\boldsymbol{\eta}) = \prod_{n=1}^{N} \eta_n^{j_n(x)-1} (1-\eta_n)^{\beta_n-1}$$

$$Q(x) = x^{p-1} \prod_{m=1}^{M} h_m(x).$$

We may generalize the previous result in a few ways:

1. For multivariate x of dimension d, if g, h, j are all linear functions mapping from $\mathbb{R}^{+,d}$ to $\mathbb{R}^{+}$, then the data augmentation is still valid.
2. The x^p term can be replaced with a polynomial function of x, as long as it's always positive for $x > 0$.
3. If the density $p(x)$ has extra factors which are not included in the general form , the strategy still works as long as we can find integral representations for those extra factors. (for example, Pólya-Gamma in negative binomial regression)

Definition

The power truncated normal distribution, PTN(p, a, b), has density function

$$p(x) = C \cdot x^{p-1} e^{-ax^2+bx}, x, p, a > 0$$

Gibbs Sampler:

$$\begin{aligned}
\tau_\ell | x, \boldsymbol{\tau}_{-\ell}, \boldsymbol{\omega}, \boldsymbol{\eta} &\sim \Gamma(g_\ell(x), 1),\ \ell = 1, 2, ..., L \\
\omega_m | x, \boldsymbol{\tau}, \boldsymbol{\omega}_{-m}, \boldsymbol{\eta} &\sim \text{ERG}(h_m(x)),\ m = 1, 2, ..., M \\
\eta_n | x, \boldsymbol{\tau}, \boldsymbol{\omega}, \boldsymbol{\eta}_{-n} &\sim \text{Beta}(j_n(x), \beta_n),\ n = 1, 2, ..., N \\
x | \boldsymbol{\tau}, \boldsymbol{\omega}, \boldsymbol{\eta} &\sim \sum_{m=0}^{M} \pi_k \cdot \text{PTN}(p + M, \tilde{a}, \tilde{b})
\end{aligned}$$

$\{\pi_m\}_{m=0}^{M}$ are proportional to the coefficients in the polynomial $Q(x)$.

$$\tilde{a} = a + \sum_{m=1}^{M} \omega_m \cdot (h'_m(0))^2,$$

$$\tilde{b} = b + \left(\sum_{\ell=1}^{L} g'_\ell(0) \cdot \log \tau_\ell \right) + \left(\sum_{m=1}^{M} h'_m(0)(\gamma - 2\omega_m h_m(0)) \right) + \left(\sum_{n=1}^{N} j'_n(0) \cdot \log \eta_n \right).$$

In summary:

- $\left(\prod_{\ell=1}^{L} \Gamma(g_\ell(x))\right)$ results in L gamma variables
- $\left(\prod_{m=1}^{M} \frac{1}{\Gamma(h_m(x))}\right)$ results in M ERG variables.
- $\left(\prod_{n=1}^{N} \frac{\Gamma(j_n(x))}{\Gamma(j_n(x)+\beta_n)}\right)$ results in N Beta variables.
- The parameters of auxiliary variables depends on g, h, j, β.
- Conditioned x, all auxiliary variables are independent.
- Conditioned on auxiliary variables, x has the density $x^{\tilde{p}-1}e^{-\tilde{a}x^2+\tilde{b}x}$.
- In many applications, g, h, j are relatively simple. Observing that $\tilde{a}$ and $\tilde{b}$ are both the sums of numerous terms, we may use normal variables to approximate them, so that the sampling procedure for $(\boldsymbol{\tau}, \boldsymbol{\omega}, \boldsymbol{\eta})$ can be skipped.

EM Algorithm

- **Expectation Step:**

$$C(x|x_{(t)}) := E_{\boldsymbol{\tau},\boldsymbol{\omega},\boldsymbol{\eta}|x_{(t)}}\left[\log p(x,\boldsymbol{\tau},\boldsymbol{\omega},\boldsymbol{\eta})\right]$$
$$= \left(\sum_{\ell=1}^{L} g_\ell(x)\psi(g_\ell(x_{(t)}))\right) + \left(\sum_{m=1}^{M} \gamma h_m(x) - h_m(x)^2 \frac{\psi(1+h_m(x_{(t)}))+\gamma}{2h_m(x_{(t)})}\right)$$
$$+ \left(\sum_{n=1}^{N} j_n(x)\psi(j_n(x_{(t)})) - \psi(j_n(x_{(t)})+\beta_n)\right) + \log Q(x) - ax^2 + bx$$

Given the linearity of g, h, j functions, it can be further simplified as

$$C(x|x_{(t)}) = \log Q(x) - \kappa_1 x^2 + \kappa_2 x$$

for some constant $\kappa_1 > 0$ and $\kappa_2 \in \mathbb{R}$, depending on $x_{(t)}$.

- **Maximization Step:**
Since $C(x|x_{(t)}) \to -\infty$ as $x \to \infty$, we conclude that $C(x|x_{(t)})$ as a function of x has a maximizer on $(0,\infty)$, which can be found numerically. Furthermore, when $h_m(0) = 0$ for all m, the unique maximizer has a closed form

$$x^* := \arg\max_{x>0} C(x|x_{(t)}) = \frac{\kappa_2 + \sqrt{\kappa_2^2 + 8\kappa_1(p+M-1)}}{4\kappa_1}.$$

Gamma Shape Inference

Gamma distribution

$$p(x|\alpha,\beta) = \frac{\beta^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\beta x}.$$

GamCon distribution of type II (Damsleth, 1975), $\xi_2(\mu,\delta)$:

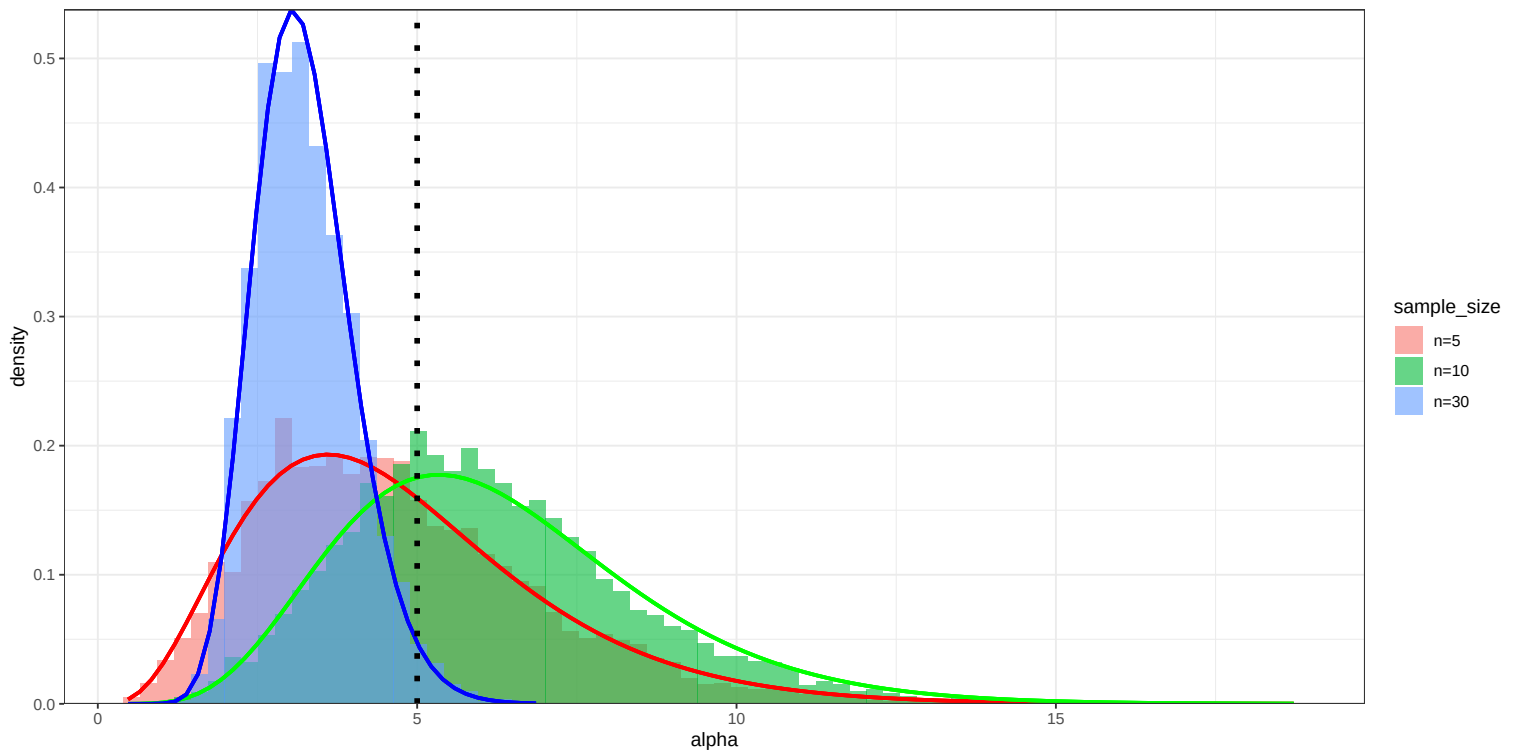
$$p(x \mid \mu,\delta) = C_{\mu,\delta} \cdot \Gamma(\delta x + 1) \left(\prod_{m=1}^{M} \frac{1}{\Gamma(x)} \right) e^{-\delta \log(\delta\mu) x}$$

Hence $L = 1$ with $g_1(x) = \delta x + 1$, $M = \delta$ with $h_m(x) = x$ and $N = 0$. $p = 1, a = 0$ and $b = -\delta \log(\delta\mu)$.

- This is the conjugate prior for the shape parameter α, when the scale parameter β is unknown. $\beta|\alpha \sim \Gamma(\delta\alpha + 1, \delta\eta)$.
- If the prior is $\xi_2(\eta/\mu, \delta)$, then the posterior is $\xi_2(\eta'/\mu', \delta')$, where

$$\begin{aligned}
\delta' &= \delta + n, \\
\eta' &= \frac{\delta}{\delta + n}\eta + \frac{n}{\delta + n}x_a, \\
\mu' &= \mu^{\frac{\delta}{\delta+n}} \cdot x_g^{\frac{n}{\delta+n}}.
\end{aligned}$$

  x_a and x_g are the arithmetical and geometrical mean respectively.
- $\delta = 0$ means non-informative prior.
- We replicate Damsleth's examples with $n = 5, 10, 30$. $\delta = 0$.



n	x_a	x_g	Method	Mean	Variance	Skewness	Kurtosis
5	7.19	6.05	Numerical Integration	4.768	5.399	0.997	4.494
			Posterior Sampling	4.779	5.630	1.305	6.148
10	5.57	5.01	Numerical Integration	6.271	5.780	0.783	3.921
			Posterior Sampling	6.101	4.857	0.634	3.427
30	5.09	4.26	Numerical Integration	3.252	0.585	0.490	3.361
			Posterior Sampling	3.250	0.605	0.537	3.155

Multinomial-Dirichlet Model

The sampling model for the count vector $\boldsymbol{n}_s$ is then a multinomial distribution given probability vector $\boldsymbol{p}_s$,

$$\boldsymbol{n}_s \mid N_s, \boldsymbol{p}_s \sim \text{Multinomial}\left(N_s, \boldsymbol{p}_s\right).$$

The prior for $\boldsymbol{p}_s$ is

$$\boldsymbol{p}_s \mid \boldsymbol{\alpha} \sim \text{Dirichlet}\left(\boldsymbol{\alpha}\right).$$

The marginal likelihood $p\left(\boldsymbol{n}_s \mid \boldsymbol{\alpha}\right)$ is

$$\prod_{s=1}^{S} p\left(\boldsymbol{n}_s \mid \boldsymbol{\alpha}\right) = \prod_{s=1}^{S} \left(\frac{\Gamma\left(\sum_{k=1}^{K} \alpha_k\right)}{\Gamma\left(\sum_{k=1}^{K} \left(n_{sk} + \alpha_k\right)\right)} \prod_{k=1}^{K} \frac{\Gamma\left(n_{sk} + \alpha_k\right)}{\Gamma\left(\alpha_k\right)} \right).$$

We now reorganize the multivariate posterior density of $\boldsymbol{\alpha}$ as

$$p(\boldsymbol{\alpha} \mid \boldsymbol{n}_1, \ldots, \boldsymbol{n}_S) = \left(\prod_{k=1}^{K} f(\alpha_k) \right) \left(\prod_{s=1}^{S} \frac{\Gamma(\sum_k \alpha_k)}{\Gamma(\sum_k \alpha_k + N_s)} \right)$$

where

$$f(\alpha_k) = \left(\prod_{s=1}^{S} \Gamma(\alpha_k + n_{sk}) \right) \left(\prod_{s=1}^{S} \frac{1}{\Gamma(\alpha_k)} \right) \alpha_k^{a_0 - 1} e^{-b_0 \alpha_k}.$$

The conditional posteriors are

$$\begin{aligned}
\tau_{sk} \mid \boldsymbol{\alpha}, \boldsymbol{\omega}, \boldsymbol{\eta} &\sim \Gamma(\alpha_k + n_{sk}), \\
\omega_{sk} \mid \boldsymbol{\alpha}, \boldsymbol{\tau}, \boldsymbol{\eta} &\sim \mathrm{ERG}(\alpha_k), \\
\eta_s \mid \boldsymbol{\alpha}, \boldsymbol{\tau}, \boldsymbol{\omega} &\sim \mathrm{Beta}\left(\sum_{k=1}^{K} \alpha_k, N_s \right), \\
\alpha_k \mid \boldsymbol{\tau}, \boldsymbol{\omega}, \boldsymbol{\eta} &\sim \mathrm{PTN}(S + a_0, a_k, b_k),
\end{aligned}$$

where $a_k = \sum_{s=1}^{S} \omega_{sk}$, $b_k = S\gamma - b_0 + \sum_{s=1}^{S} \log(\tau_{sk} \cdot \eta_s)$.

Simulation Study

- (A) heterogeneous $\alpha_k = k/K$ and (B) homogeneous $\alpha_k = 1/K$.
- $a_0 = b_0/K$, so that the prior expectations are all equal to $1/K$.
- $N_s = 500$ for $s = 1, 2, ..., S$.
- $S \in \{100, 1000\}, K \in \{10, 50\}$ and $b_0 \in \{0.1, 1, 5\}$.
- 4 MCMC algorithms in comparison:
  - DA: Gibbs sampler which iteratively samples $\boldsymbol{\tau}, \boldsymbol{\omega}, \boldsymbol{\eta}$ and $\boldsymbol{\alpha}$.
  - DA-N: replace the parameters in the conditional posterior of $\boldsymbol{\alpha}$ with approximating normal variables and skip the sampling of $\boldsymbol{\tau}, \boldsymbol{\omega}, \boldsymbol{\eta}$ in DA.
  - DA-E: replace the parameters in the conditional posterior of $\boldsymbol{\alpha}$ with corresponding expectations and skip the sampling of $\boldsymbol{\tau}, \boldsymbol{\omega}, \boldsymbol{\eta}$ in DA.
  - MH: random-walk Metropolis-Hasting sampler.
- For each combination of (b_0, S, K), we run the simulation for 50 times (the first 200 samples are dropped each time).

Metrics:

- Root Mean Square Error:

$$\text{RMSE} = \sqrt{\frac{1}{TK}\sum_{t=1}^{T}\|\boldsymbol{\alpha}^{(t)} - \boldsymbol{\alpha}\|^2}.$$

  $T = 500$ is the posterior sample size.

- Effective sample size ratio (ESSR):

$$\text{ESSR} = \frac{1}{K}\sum_{k=1}^{K}\frac{\lambda_k^2}{\sigma_k^2},$$

  which measures the serial correlation between posterior samples. λ^2 is the sample variance and σ^2 is the estimate of the spectral density at frequency zero.

- Algorithm running time in seconds

Table 1: Homogeneous Setting

		$b_0 = 0.1$				$b_0 = 1$				$b_0 = 5$			
		Root Mean Square Error ($\times 10^3$)											
S	K	DA	DA-N	DA-E	MH	DA	DA-N	DA-E	MH	DA	DA-N	DA-E	MH
100	10	20.93	20.95	18.95	20.56	20.25	20.36	18.31	19.99	20.22	20.25	18.38	20.19
100	50	7.78	8.03	6.93	5.88	7.71	7.95	6.86	6.03	7.66	7.95	6.84	5.95
1000	10	6.54	6.53	5.95	7.13	6.58	6.54	6.01	7.26	6.38	6.36	5.80	6.87
1000	50	2.49	2.51	2.19	2.48	2.49	2.51	2.20	2.50	2.48	2.50	2.18	2.46
		Effective Sample Size Ratio											
S	K	DA	DA-N	DA-E	MH	DA	DA-N	DA-E	MH	DA	DA-N	DA-E	MH
100	10	0.32	0.33	0.32	0.04	0.32	0.32	0.32	0.04	0.32	0.32	0.32	0.04
100	50	0.08	0.08	0.07	0.01	0.08	0.08	0.07	0.01	0.08	0.08	0.07	0.01
1000	10	0.32	0.32	0.32	0.01	0.33	0.33	0.32	0.01	0.32	0.32	0.32	0.01
1000	50	0.08	0.08	0.07	0.01	0.08	0.07	0.07	0.01	0.08	0.08	0.07	0.01
		Running Time											
S	K	DA	DA-N	DA-E	MH	DA	DA-N	DA-E	MH	DA	DA-N	DA-E	MH
100	10	0.62	0.03	0.02	0.04	0.60	0.03	0.02	0.04	0.63	0.03	0.02	0.04
100	50	3.16	0.07	0.03	0.05	3.01	0.07	0.03	0.05	2.99	0.07	0.03	0.05
1000	10	4.01	0.10	0.03	0.17	3.76	0.09	0.03	0.17	3.74	0.09	0.03	0.17
1000	50	19.85	0.53	0.11	0.46	18.47	0.52	0.11	0.44	18.50	0.52	0.11	0.44

Table 2: Heterogeneous Setting

		$b_0 = 0.1$				$b_0 = 1$				$b_0 = 5$			
		Root Mean Square Error ($\times 10^3$)											
S	K	DA	DA-N	DA-E	MH	DA	DA-N	DA-E	MH	DA	DA-N	DA-E	MH
100	10	82.76	82.25	76.54	82.34	82.56	81.90	77.33	83.49	81.95	82.11	78.46	86.96
100	50	81.50	81.41	75.83	553.05	78.53	78.39	73.47	553.83	81.99	81.98	79.15	553.38
1000	10	26.11	26.00	24.50	41.00	25.76	25.61	24.18	41.08	25.88	25.71	24.25	41.39
1000	50	24.80	24.79	23.19	553.35	25.22	25.11	23.66	553.30	25.26	25.21	23.71	552.86
		Effective Sample Size Ratio											
S	K	DA	DA-N	DA-E	MH	DA	DA-N	DA-E	MH	DA	DA-N	DA-E	MH
100	10	0.47	0.49	0.50	0.03	0.48	0.51	0.49	0.03	0.49	0.51	0.51	0.03
100	50	0.44	0.45	0.44	0.01	0.44	0.45	0.44	0.01	0.45	0.46	0.46	0.01
1000	10	0.48	0.50	0.50	0.02	0.47	0.50	0.50	0.01	0.47	0.50	0.50	0.01
1000	50	0.44	0.44	0.44	0.01	0.44	0.45	0.45	0.01	0.44	0.44	0.45	0.01
		Running Time											
S	K	DA	DA-N	DA-E	MH	DA	DA-N	DA-E	MH	DA	DA-N	DA-E	MH
100	10	0.74	0.09	0.08	0.10	0.68	0.09	0.08	0.10	0.68	0.09	0.08	0.10
100	50	3.23	0.13	0.10	0.58	3.07	0.13	0.09	0.58	3.08	0.13	0.10	0.58
1000	10	3.92	0.09	0.05	0.20	3.77	0.09	0.05	0.20	3.76	0.09	0.05	0.19
1000	50	19.44	0.42	0.13	0.97	18.53	0.41	0.13	0.95	18.54	0.41	0.13	0.95

Discussion

- The class of ERG distributions are developed as mixing distributions for models with gamma functions. This adds to the literature on normal variance-mean mixtures by showing that they extend to a wide class of applications.
- Our ensuing data augmentation strategy facilitates full posterior inference for parameters in models which were hitherto hard to provide inference.
- We believe that the computational strategies developed here will provide the foundation for new and richly structured hierarchical gamma models.
- Avenues for future research: regularized scale allocation models, multivariate ERG theory, etc.

Thanks!